ASTDS Quiz 3

March 20, 2025

It's a closed-book and closed-note examination. Answer all questions.

Time: 20 minutes

1. Suppose in a random sample of 20 DAU students, 2 are found to spend more than 6 hours on average daily on social networks. Suppose θ denotes the probability of spending more than 6 hours for a randomly chosen student.
   (a) Write down the maximum likelihood estimate of θ.
   (b) Find the posterior distribution of θ using the above data assuming Beta (a, b) as a prior.
   (c) Find the posterior mean and write it down as a convex combination of the Max likelihood estimate and the mean of the prior distribution.
   (d) Consider the following choices of the prior: (a, b) is equal to $(0.1, 0.9)$ and $(10, 90)$. Discuss in which case prior is dominating the Bayes' estimate considering it as the convex combination stated above.

(Points $5 + 3 + 8 + 4 = 20$)

(Beta (α, β) density is given by $f(x|\alpha, \beta) \propto x^{\alpha-1}(1-x)^{\beta-1}, 0 < x < 1$.

Mean $= \alpha/(\alpha + \beta)$)

(a): Maximum Likelihood Estimate of θ

We assume that the number of students spending more than 6 hours on social networks follows a binomial distribution:

$$X \sim \text{Bin}(n = 20, \theta)$$

The likelihood function is:

$$L(\theta) = P(X = 2|n = 20, \theta) = \binom{20}{2}\theta^2(1-\theta)^{18}$$

Taking the log-likelihood:

$$\log L(\theta) = \log \binom{20}{2} + 2\log\theta + 18\log(1-\theta)$$

Differentiating and solving for θ:

$$\frac{d}{d\theta}\left[2\log\theta + 18\log(1-\theta)\right] = \frac{2}{\theta} - \frac{18}{1-\theta} = 0$$

Solving,

$$2(1-\theta) = 18\theta$$

$$2 = 20\theta$$

$$\hat{\theta} = \frac{2}{20} = 0.1$$

Thus, the MLE estimate is:

$$\hat{\theta} = 0.1$$

(b): Posterior Distribution of θ

Assuming a Beta prior:

$$\theta \sim \text{Beta}(a, b)$$

The prior density is:

$$p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$$

Using Bayes' theorem:

$$p(\theta|X) \propto L(\theta)p(\theta)$$

$$\propto \theta^2(1-\theta)^{18} \times \theta^{a-1}(1-\theta)^{b-1}$$

$$\propto \theta^{a+1}(1-\theta)^{b+17}$$

Thus, the posterior distribution is:

$$\theta|X \sim \text{Beta}(a+2, b+18)$$

(c): Posterior Mean as a Convex Combination

The mean of a Beta(α, β) distribution is:

$$E[\theta|X] = \frac{a+2}{a+b+20}$$

Expressing it as a **convex combination** of the MLE estimate $\hat{\theta}$ and the prior mean:

$$E[\theta|X] = w\hat{\theta} + (1-w)\frac{a}{a+b}$$

where

$$w = \frac{20}{a+b+20}$$

Thus,

$$E[\theta|X] = \frac{20}{a+b+20}(0.1) + \frac{a+b}{a+b+20}\frac{a}{a+b}$$

(d): Prior Influence on Posterior Estimate

Case 1: $(a, b) = (0.1, 0.9)$

$$w = \frac{0.1+0.9}{0.1+0.9+20} = \frac{1}{21} \approx 0.0476$$

The posterior mean is dominated by the MLE (weight $1-w \approx 0.9524$).

Case 2: $(a, b) = (10, 90)$

$$w = \frac{10+90}{10+90+20} = \frac{100}{120} \approx 0.8333$$

The posterior mean is dominated by the prior (weight $w \approx 0.8333$).

Conclusion: - When the prior is weak (e.g., 0.1, 0.9), the posterior estimate is dominated by the MLE. When the prior is strong (e.g., 10, 90), it dominates the Bayes estimate, leading to an estimate close to the prior mean.